

C:\Users\ziyou\AppData\Local\Temp\ipykernel_15192\2635408567.py:195:

MatrixRankWarning: Matrix is exactly singular

```
U = spla.spsolve(A_mat, b)
```

Solution computed successfully.

Grid size: 41x41

Maximum ux: nan m

Maximum uy: nan m

Maximum displacement magnitude: nan m

Maximum σ_{xx} : nan Pa

Maximum σ_{yy} : nan Pa

Maximum σ_{xy} : nan Pa